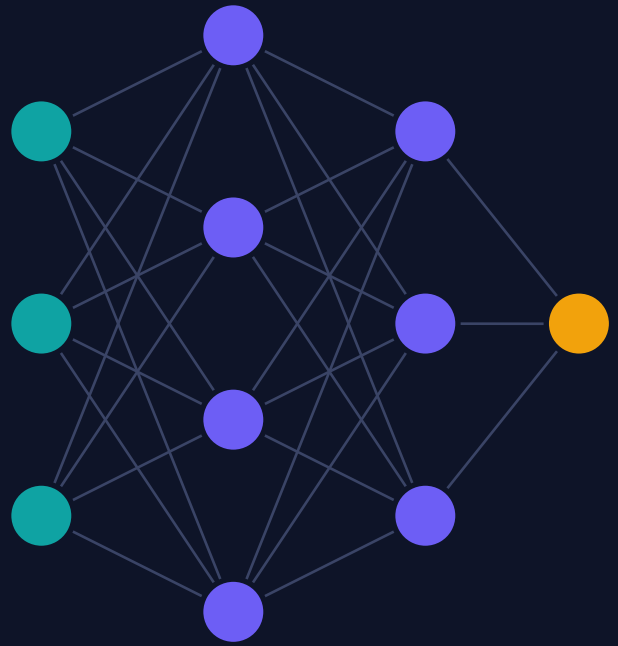
A Short History of AI

From hand-written rules to machines that learn from data



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Traditional programming vs. machine learning

Traditional programming



WHO WRITES THE LOGIC

A programmer writes explicit rules

HOW IT ADAPTS

Someone changes the code when needs change

HOW IT'S BUILT

Linear and predictable: implement, then debug

Machine learning



WHO WRITES THE LOGIC

The model learns the rules from example data

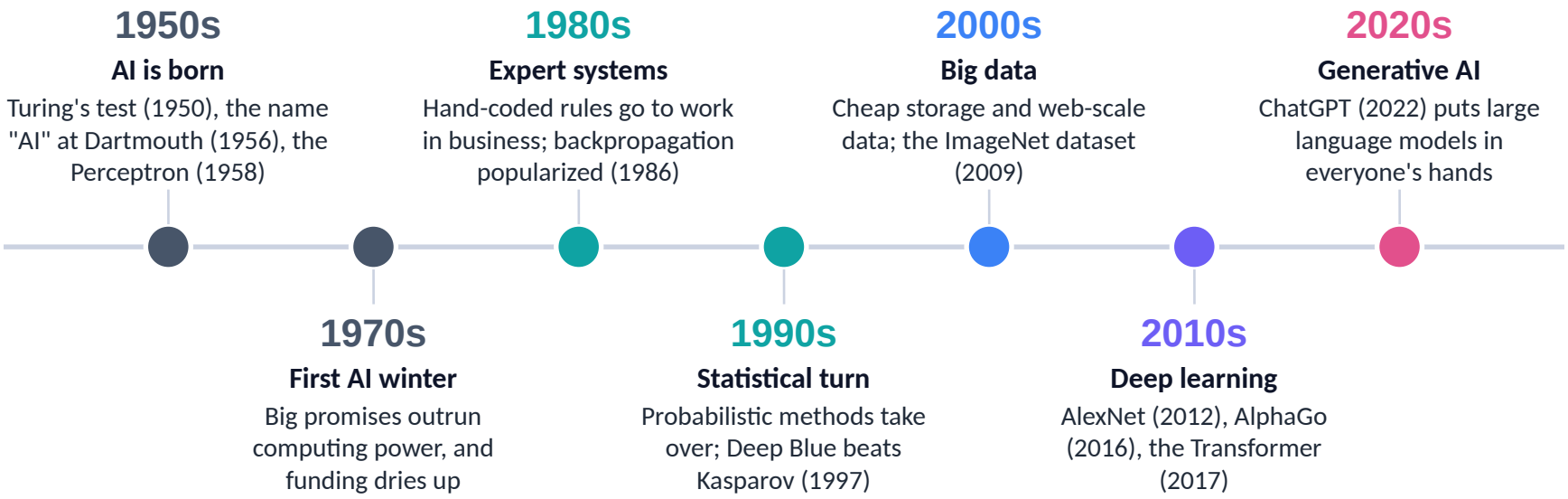
HOW IT ADAPTS

Retrain on new data to handle new scenarios

HOW IT'S BUILT

Iterative: train, evaluate, tune, repeat

Seven decades of AI, in waves



Today, the hard questions are about data

How models get their knowledge

1. **Experts write rules**
   1980s
2. **People label data**
   2000s
3. **Pre-trained models**
   2018 onward, then fine-tuned

How much data is enough?

More data helps, but each extra example costs time and money.

Who labels it, and how good are the labels?

Annotation quality sets the ceiling for model quality.

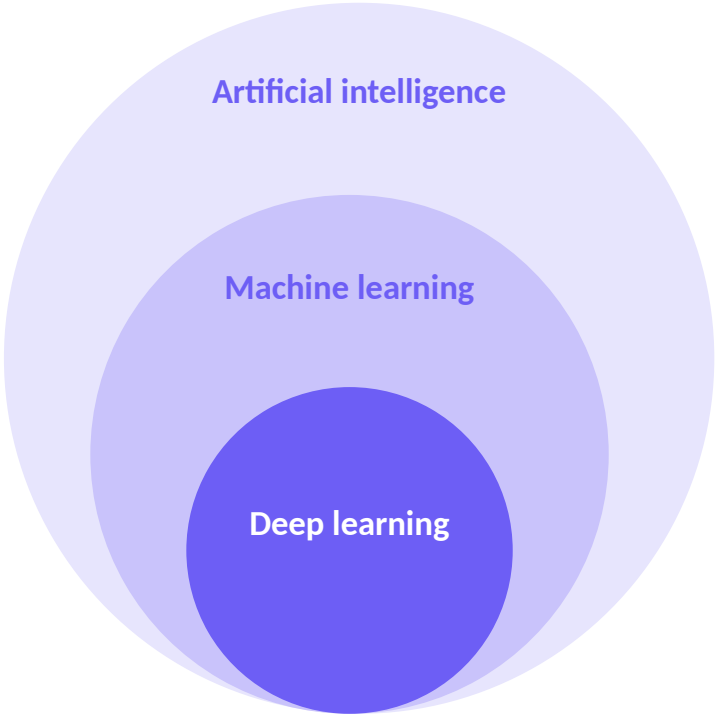
Can we skip paying for labels?

Synthetic data and pre-trained models can cut the cost.

What should humans review next?

Model uncertainty points to the most useful examples.

AI, machine learning and deep learning



Supervised learning
Learns from labeled examples: spam filters, price forecasts

Unsupervised learning
Finds structure in unlabeled data: customer segments, anomalies

Reinforcement learning
Learns by trial and reward: game play, robotics

Inside deep learning
Early layers spot edges; later layers combine them into shapes, then faces

Where AI already changes how businesses work



Customer experience

Chat assistants and faster support routing



Financial analytics

Forecasting, risk scoring and reporting



Security

Spotting threats in network and log data



Recommendations

Personalized products, content and offers



Predictive analytics

Churn, maintenance and demand signals



Product demand

Inventory and supply planning



Fraud detection

Flagging unusual transactions in real time



Content

Generating drafts and moderating posts

Key takeaways

01

From rules to learning

Programs no longer need every rule written by hand. Models infer them from data.

02

Progress comes in waves

Each leap combined more compute, more data and better algorithms.

03

Data is the edge

Clean, well-modelled, trustworthy data is what makes AI work, which is where data engineering comes in.

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Thanks for reading. I'd be glad to connect.